

Supplementary Material for “Adapting Before Contact: Contextual Memory for Frozen World Models”

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This supplement contains the proofs of all formal results, additional experimental results, implementation details, and the timeline of the hardware deployment. Equations, figures and tables numbered without the prefix “S” refer to the main paper.

A Overview of the Formal Results

Table S1 lists every formal result, where it is stated, and what it assumes. Each proof in Appendix B first restates the result, then gives the idea in a few sentences, and then proceeds in short numbered steps. Table S2 collects the notation.

Result	Statement in one line	Key assumptions
Observation 1 (first-contact initialization)	Error-driven, episode-reset adapters predict with the frozen model at the first contact.	No feedback before commitment
Theorem 1 (memory error)	Memory error $\leq$ noise + prior mismatch + contamination; contamination is bounded by the filter’s log loss.	Assumption 1; predictive weights
Remark 1 (filtered weights)	Posterior-weighted updates converge to a biased value with shrinking intervals.	Fixed scoring models
Proposition 1 (first-contact separation)	Cumulative first-contact bias $\leq \sum \epsilon_e + \Delta_{\max} \times$ episode log loss; reset methods pay $\geq N_{\text{shift}} \Delta_{\min}$.	Closed library
Corollary 1 (novel branch)	Extends Proposition 1 to a mixture that includes the new-context hypothesis.	Bound on novel-branch error
Corollary 2 (entropy reference)	A KT predictor has log loss $\leq N \hat{H}(C \mid Y) + O(K \mathcal{Y} \log N)$.	Revealed labels
Lemma 1, Corollary 3 (decision gap)	Prediction error along candidate rollouts bounds the committed-action gap.	Lipschitz dynamics, costs
Proposition 2 (composition)	Additive transfer to an unseen pair iff the observed-combination graph connects it; interactions leave a variance floor.	Exact stored means
Proposition 3 (value of information)	VOI ≥ 0 , and VOI = 0 when one action is optimal for all plausible states.	Fixed state and action set
Proposition 4 (latent safety)	Violations $\leq \delta_s T + K_T(\bar{S} + 2\gamma)/\gamma$ for any context and cue sequence.	Bounded scores, exact filter
Proposition 5 (recognition time)	Expected steps to recognition = $(\ell_* - \ell_0 + \mathbb{E}[O_T])/D$.	i.i.d. increments

Table S1: Formal results and their assumptions.

Symbol	Meaning	Symbol	Meaning
z_t, a_t	latent state, action	f	frozen predictor
c_e, c_t	true context (episode, step)	μ_c	mean residual of context c
U, ϕ	correction subspace, features	$v_t = U^\top r_t$	projected residual
$w_{c,i}, \hat{w}_{c,i,t}$	true / estimated weights (row i)	m_c	latent correction of memory c
$q_t(c)$	belief <i>before</i> observing v_t	$\pi_t^+(c)$	belief <i>after</i> observing v_t
$\pi_e(c)$	pre-contact belief in episode e	y_e	visual cue
$\omega_{c,i,t}$	update weight $q_t(c)/\sigma_i^2$	$V_{c,i,t}$	weighted design matrix
$P_{c,i}, \nu_{c,i,t}$	prior precision, prior mean	$\Delta_{\max}$	largest residual difference
h, L_h	latent safety function, Lipschitz constant	$b_{c,t}$	safety margin of memory c
$s_{c,t}$	calibration score of memory c	δ_s, γ	target miss rate, step size

Table S2: Notation used in the proofs. We write $\|x\|_A = \sqrt{x^\top A x}$, and $\mathcal{F}_{t-1}$ for the history up to and including (z_t, a_t) , the cue and the true context c_t .

B Proofs

B.1 First-contact initialization (Observation 1)

Idea. A method that only reacts to errors cannot change before the first error arrives.

Proof. Let $S_{e,t}$ be the adaptive state and suppose $S_{e,t+1} = S_{e,t}$ whenever no prediction-error feedback is received at step t . By assumption no such feedback is available before τ_e , so by induction $S_{e,t} = S_{e,0}$ for all $t \leq \tau_e$. The prediction is a function of the state and the input (z, a) only, hence the prediction at τ_e equals the prediction at the start of the episode. If $S_{e,0}$ is the frozen model, this prediction is $f(z, a)$, and its bias is $\|f + \mu_{c_e} - f\| = \|\mu_{c_e}\|$. $\square$

B.2 Uniform memory error (Theorem 1)

Theorem 1 (restated). *Under Assumption 1, with predictive weights and with probability at least $1 - \delta_c$, simultaneously for all t , ϕ and i , $|\phi^\top(\hat{w}_{c,i,t} - w_{c,i})| \leq \beta_{c,i,t} \|\phi\|_{V_{c,i,t}^{-1}}$ with $\beta_{c,i,t}$ the sum of a noise, a prior-mismatch and a contamination term, and the total contamination is bounded by the log loss of the context filter.*

Idea. The estimate $\hat{w}$ is a weighted ridge regression. Its error splits into three parts: noise from the current context, bias from data that actually came from other contexts, and bias from the prior mean. Each part is bounded separately in the norm induced by V_t . Only the noise part needs probability, and it can be bounded with a martingale inequality because the weights are chosen before the noise is drawn.

Proof. Fix c and a row i and drop both from the notation. Let $\omega_s = q_s(c)/\sigma^2(z_s, a_s)$ and define

$$x_s = \sqrt{\omega_s} \phi_s, \quad \eta_s = \sqrt{\omega_s} \xi_s, \quad d_s = \sqrt{\omega_s} \phi_s^\top (w_{c_s} - w_c),$$

so that, by Assumption 1, $\sqrt{\omega_s} v_s = x_s^\top w_c + d_s + \eta_s$.

Step 1 (decomposition). Substituting this into Eq. (6) and using $V_t = P + \sum_{s < t} x_s x_s^\top$ gives

$$\hat{w}_t - w_c = \underbrace{V_t^{-1} \sum_{s < t} x_s \eta_s}_{\text{noise}} + \underbrace{V_t^{-1} \sum_{s < t} x_s d_s}_{\text{contamination}} + \underbrace{V_t^{-1} P(\nu_t - w_c)}_{\text{prior}}.$$

For any ϕ and u , the Cauchy–Schwarz inequality in the V_t^{-1} inner product gives $|\phi^\top V_t^{-1} u| \leq \|\phi\|_{V_t^{-1}} \|u\|_{V_t^{-1}}$. It remains to bound $\|u\|_{V_t^{-1}}$ for the three vectors $u = \sum_s x_s \eta_s$, $\sum_s x_s d_s$ and $P(\nu_t - w_c)$.

Step 2 (noise). The vector x_s is $\mathcal{F}_{s-1}$ -measurable: ϕ_s and $\sigma(z_s, a_s)$ depend only on (z_s, a_s) , and $q_s(c)$ depends only on data before v_s and on the cue. Given $\mathcal{F}_{s-1}$, η_s is $\sqrt{q_s(c)}$ -sub-Gaussian and therefore 1-sub-Gaussian. The self-normalized bound for vector-valued martingales [1, Thm. 1], with regularizer P and confidence δ_c/k , yields simultaneously for all t

$$\left\| \sum_{s < t} x_s \eta_s \right\|_{V_t^{-1}}^2 \leq 2 \log \frac{k}{\delta_c} + \log \frac{\det V_t}{\det P}.$$

This is the only step that uses predictability. With the filtered weight $\pi_s^+(c)$, x_s would depend on ξ_s , the sum would no longer be a martingale transform, and the inequality would not apply.

Step 3 (contamination). Let X be the matrix with rows $x_s^\top$ and $d = (d_s)_{s < t}$. Since $V_t \succeq X^\top X$, the matrix $X V_t^{-1} X^\top$ has eigenvalues at most one, so $\|X^\top d\|_{V_t^{-1}}^2 = d^\top X V_t^{-1} X^\top d \leq \|d\|^2 = B_t^2$. This holds for every realization and every choice of weights. Since $d_s = 0$ when $c_s = c$ and $|d_s| \leq \sqrt{\omega_s} \Delta_{\max}$ otherwise, $B_t^2 \leq \Delta_{\max}^2 M_t$.

Step 4 (prior). Since $V_t \succeq P$, we have $P V_t^{-1} P \preceq P$ and hence $\|P(\nu_t - w_c)\|_{V_t^{-1}}^2 \leq \|\nu_t - w_c\|_P^2$. This also holds for every realization, so the prior mean may be recomputed from past data.

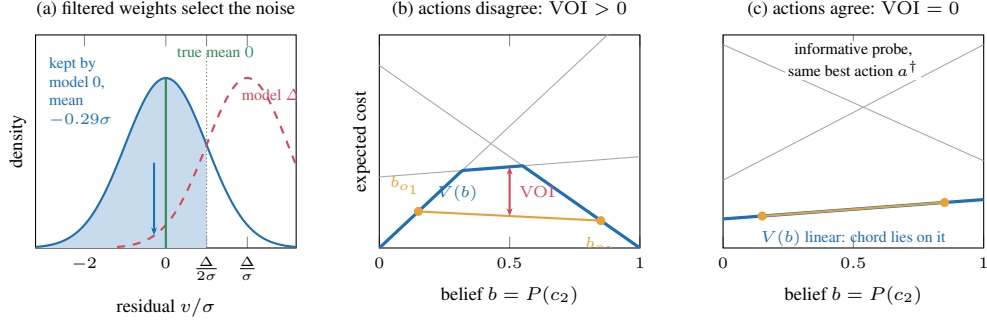


Figure S1: **Two mechanisms behind the analysis** (analytic illustrations). (a) **Remark 1**: when each residual is assigned by the filtered posterior, model 0 keeps only residuals left of $\Delta/2$ (shaded). Its mean converges to -0.29σ instead of 0. (b,c) **Proposition 3**: $V(b)$ is the lower envelope of the action costs and is concave. A probe splits the belief into b_{o_1} and b_{o_2} . The value of information is the gap between $V(b)$ and the chord. When one action is best for every plausible context (c), V is linear and the gap vanishes, although the probe is informative.

Step 5 (combination). Adding the three bounds and taking a union bound over the k rows proves the first claim. For Eq. (9), $\sum_c M_{c,T} = \sum_{t < T} \sum_{c \neq c_t} q_t(c)/\sigma_t^2 \leq \sigma_{\min}^{-2} \sum_{t < T} (1 - q_t(c_t))$, and $1 - x \leq -\log x$ for $x \in (0, 1]$. $\square$

Remark 1 (Filtered weights bias the memory). Let the residuals of context 0 be $v \sim \mathcal{N}(0, \sigma^2)$, and let two fixed models with means 0 and $\Delta > 0$ have equal prior probability. If each residual is assigned to the model with the larger filtered probability, model 0 receives it exactly when $v < \Delta/2$. Its estimate converges to $\mathbb{E}[v \mid v < \Delta/2] = -\sigma \varphi(\Delta/2\sigma)/\Phi(\Delta/2\sigma)$, which is about -0.29σ at $\Delta = 2\sigma$, while its interval shrinks as $\sigma/\sqrt{n}$ and eventually excludes the true value 0.

Derivation. With equal priors and variances, the filtered probability of model 0 is larger exactly when $|v| < |v - \Delta|$, that is, when $v < \Delta/2$. The mean of a standard normal truncated above at a is $-\varphi(a)/\Phi(a)$; with $a = \Delta/2\sigma = 1$ this gives $-\sigma\varphi(1)/\Phi(1) \approx -0.288\sigma$. By the law of large numbers the sample mean of the selected residuals converges to this value, while its standard error is $O(\sigma/\sqrt{n})$. Predictive weights do not depend on v and avoid the selection (Fig. S1a).

B.3 First-contact separation (**Proposition 1**) and its extensions

Proposition 1 (restated). *If the library contains every occurring context and $\|m_c - \mu_c\| \leq \epsilon_e$ on the support of π_e , then $b_e \leq \epsilon_e + (1 - \pi_e(c_e))\Delta_{\max}$ and $\sum_e b_e \leq \sum_e \epsilon_e + \Delta_{\max} \sum_e -\log \pi_e(c_e)$.*

Idea. The mixture prediction is correct up to ϵ_e on the true memory, and every unit of belief placed on a wrong memory costs at most $\Delta_{\max}$.

Proof. Step 1. Write the mixture prediction as $\sum_c \pi(c)m_c$ and add and subtract $\sum_c \pi(c)\mu_c$:

$$\left\| \sum_c \pi(c)m_c - \mu_{c_e} \right\| \leq \sum_c \pi(c)\|m_c - \mu_c\| + \left\| \sum_{c \neq c_e} \pi(c)(\mu_c - \mu_{c_e}) \right\|.$$

Step 2. The first term is at most ϵ_e because the weights sum to one. The second is at most $\sum_{c \neq c_e} \pi(c)\Delta_{\max} = (1 - \pi(c_e))\Delta_{\max}$.

Step 3. Summing over episodes and using $1 - x \leq -\log x$ gives the cumulative bound. The lower bound for reset methods follows directly from **Observation 1**. $\square$

Corollary 1 (Mixture with a novel-context branch). *Suppose the true context c_e is stored, the stored memories satisfy $\|m_c - \mu_c\| \leq \epsilon_e$, and the new-context hypothesis carries belief $p_\star$ with conditional mean $m_\star$. Then*

$$b_e \leq (1 - p_\star)\epsilon_e + (1 - p_\star - \pi_e(c_e))\Delta_{\max} + p_\star\|m_\star(x_e) - \mu_{c_e}(x_e)\|.$$

Proof. Split the mixture into the stored memories, with total mass $1 - p_*$, and the new-context branch. On the stored part, the argument above applies with the wrong memories carrying mass $1 - p_* - \pi_e(c_e)$. The new-context branch contributes at most $p_* \|m_* - \mu_{c_e}\|$ by the triangle inequality. $\square$

Without a bound on $\|m_* - \mu_{c_e}\|$, the closed-library guarantee does not cover the full mixture. A genuinely new condition requires an acquisition phase or a valid transferred prior.

Corollary 2 (Finite-alphabet entropy reference). *Suppose there are K contexts and a finite cue alphabet $\mathcal{Y}$, and each episode’s context is revealed after its prediction. For each cue value, predict with the Krichevsky–Trofimov estimator $\pi_e^{\text{KT}}(c \mid y) = (n_{<e}(c, y) + \frac{1}{2}) / (n_{<e}(y) + \frac{K}{2})$. Then for every context–cue sequence, $\sum_e -\log \pi_e^{\text{KT}}(c_e \mid y_e) \leq N \hat{H}(C \mid Y) + O(K|\mathcal{Y}| \log(N+1))$.*

Proof. Split the sequence by cue value. For the N_y episodes with cue y , the Krichevsky–Trofimov estimator has log loss at most $N_y \hat{H}(C \mid Y = y) + \frac{K-1}{2} \log N_y + O(K)$ [2]. Summing over y gives $\sum_y N_y \hat{H}(C \mid Y = y) = N \hat{H}(C \mid Y)$ plus at most $|\mathcal{Y}|(\frac{K-1}{2} \log(N+1) + O(K))$. $\square$

CIRA does not receive labels, so its inferred beliefs do not automatically inherit this bound. We therefore report the measured log loss of CIRA’s beliefs alongside $\hat{H}(C \mid Y)$.

B.4 From prediction error to the committed decision

Lemma 1 (Simulation bound). *Fix an action sequence a , and let z_t and $\hat{z}_t$ be the noise-free trajectories of the true map F and a model $\hat{F}$ from the same initial state. If F is L -Lipschitz in the state and the stage and terminal costs are L_ℓ -Lipschitz, then $|J(a) - \hat{J}(a)| \leq L_\ell C_H \sum_{t < H} \|F(\hat{z}_t, a_t) - \hat{F}(\hat{z}_t, a_t)\|$ with $C_H = \sum_{j < H} L^j$.*

Idea. Errors made at one step are propagated, and at most amplified by L , at each later step.

Proof. Let $E_t = \|z_t - \hat{z}_t\|$ and $\text{err}_t = \|F(\hat{z}_t, a_t) - \hat{F}(\hat{z}_t, a_t)\|$. Adding and subtracting $F(\hat{z}_t, a_t)$ gives $E_{t+1} \leq L E_t + \text{err}_t$. Unrolling from $E_0 = 0$ yields $E_t \leq \sum_{s < t} L^{t-1-s} \text{err}_s$. The cost difference is at most $L_\ell \sum_t E_t$, and exchanging the order of summation gives the claim. $\square$

Corollary 3 (First-contact decision gap). *Let $\mathcal{A}_e$ be a finite candidate set, $\hat{J}_e(a) = \sum_c \pi_e(c) \hat{J}_{e,c}(a)$ with $\hat{J}_{e,c}$ rolling out $f + m_c$, and let $\hat{a}_e$ minimize $\hat{J}_e$ up to η_e . If $\|m_c(x) - \mu_c(x)\| \leq \bar{\epsilon}_e$ for every supported context along every candidate rollout, then*

$$J_e(\hat{a}_e) - \min_{a \in \mathcal{A}_e} J_e(a) \leq 2L_\ell C_H H [\bar{\epsilon}_e + (1 - \pi_e(c_e)) \Delta_{\max}] + \eta_e,$$

and summing over episodes replaces $1 - \pi_e(c_e)$ by $-\log \pi_e(c_e)$ as in [Proposition 1](#).

Proof. *Step 1.* By [Lemma 1](#) applied to each memory, and because U has orthonormal columns, the per-step model error of memory c is at most $\bar{\epsilon}_e + \mathbf{1}[c \neq c_e] \Delta_{\max}$. Averaging over the belief gives $|\hat{J}_e(a) - J_e(a)| \leq L_\ell C_H H (\bar{\epsilon}_e + (1 - \pi_e(c_e)) \Delta_{\max})$ for every candidate a .

Step 2. Let a^* minimize J_e over $\mathcal{A}_e$. Then $J_e(\hat{a}_e) - J_e(a^*) \leq [J_e(\hat{a}_e) - \hat{J}_e(\hat{a}_e)] + [\hat{J}_e(a^*) - J_e(a^*)] + \eta_e$, and both brackets are bounded by Step 1. $\square$

The second bracket is evaluated on the rollout of the unknown best action, which is why the memory error must be bounded uniformly over the candidate set; [Theorem 1](#) provides such a uniform bound.

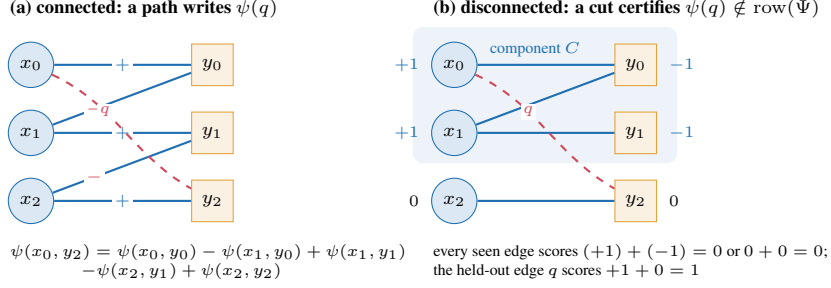


Figure S2: **Why connectivity decides additive transfer** (proof of Proposition 2). Circles are surface values and squares are load values; solid edges are seen combinations and the dashed edge is the query q . (a) If a path of seen edges joins q 's two values, alternating signs along the path cancel every interior indicator, so $\psi(q)$ is a combination of seen rows and its additive part is determined. (b) If no path exists, assigning $+1$ to surfaces and -1 to loads inside one component gives a vector orthogonal to every seen row but not to $\psi(q)$. The additive part of q therefore remains uncertain.

B.5 Identifiability of unseen combinations (Proposition 2)

Proposition 2 (restated). *Under the factored prior with exact observations on Ω : (i) if $\kappa_\times = 0$, the conditional variance of $g(q)$ vanishes if and only if the indicators of q lie in the span of those of Ω , which for two factors means that q 's values are connected in the observation graph; (ii) if $\kappa_\times > 0$, the conditional variance is at least $\kappa_\times^2 \phi^\top \Sigma_0 \phi$.*

Idea. Without interactions, $g(q)$ is a linear function of per-factor effects. It is determined by the observations exactly when its indicator vector can be written as a signed combination of observed indicator vectors, which in the bipartite graph corresponds to a path. An interaction specific to q is independent of everything observed and can never be learned from other combinations. Figure S2 illustrates both directions of the argument.

Proof. Under Eq. (4), $g(c) = \psi(c)^\top \beta + \xi_c$, where $\psi(c)$ stacks a constant and the one-hot indicators of $c^1, \dots, c^F$, β is Gaussian with positive definite diagonal covariance D , and the $\xi_c \sim \mathcal{N}(0, \kappa_\times^2 \phi^\top \Sigma_0 \phi)$ are independent of β and of each other. Let Ψ have rows $\psi(\omega)^\top$ for $\omega \in \Omega$.

Step 1 (span criterion). With $\kappa_\times = 0$ the observations are $\Psi\beta$. For Gaussian β , the conditional variance of $\psi(q)^\top \beta$ equals the squared distance from $D^{1/2}\psi(q)$ to the range of $D^{1/2}\Psi^\top$, which is zero exactly when $\psi(q)$ lies in the row space of Ψ .

Step 2 (path $\Rightarrow$ span). For two factors, suppose a path $q^1 = x_0, y_0, x_1, y_1, \dots, x_m, y_m = q^2$ alternates between factor values along observed edges (x_j, y_j) and (x_{j+1}, y_j) . Then

$$\psi(q) = \sum_{j=0}^m \psi(x_j, y_j) - \sum_{j=0}^{m-1} \psi(x_{j+1}, y_j),$$

because the constant appears $m+1$ times with a plus sign and m times with a minus sign, and every interior indicator appears once with each sign (Fig. S2a).

Step 3 (no path $\Rightarrow$ no span). If q^1 and q^2 lie in different components, let C be the component of q^1 and define u with constant entry 0, entry $+1$ for factor-1 values in C , entry -1 for factor-2 values in C , and 0 elsewhere. Every observed edge lies entirely inside or entirely outside C , so $u^\top \psi(\omega) = 0$ for all $\omega \in \Omega$, while $u^\top \psi(q) = 1$. Hence $\psi(q)$ is not in the row space (Fig. S2b).

Step 4 (interaction floor). If $\kappa_\times > 0$, the term ξ_q is independent of β and of all observations, so the conditional variance of $\psi(q)^\top \beta + \xi_q$ is the sum of the conditional variance of $\psi(q)^\top \beta$ and $\kappa_\times^2 \phi^\top \Sigma_0 \phi$. $\square$

Exact conditional variance. With $\chi = \phi^\top \Sigma_0 \phi$ and the Gram matrix $K_\Omega = [k_C(c, c')]_{c, c' \in \Omega}$, Gaussian conditioning gives $\text{Var}(g(q) \mid g(\Omega)) = \chi s_q$ with $s_q = k_C(q, q) - k_C(q, \Omega) K_\Omega^\dagger k_C(\Omega, q)$, where

$\dagger$ denotes the Moore–Penrose inverse. The pseudoinverse is needed because the additive kernel can be singular even when all variance parameters are positive. We compute s_q for every held-out combination before evaluation and use it as the predicted first-contact uncertainty.

B.6 Value of information (Proposition 3)

Proposition 3 (Value of information under action agreement). *For a fixed state and a finite action set, $\text{VOI}(p) \geq 0$ for every probe. If a single action minimizes J_θ for almost every θ under b , then $\text{VOI}(p) = 0$ for every probe, even though the mutual information $I(\theta; o \mid p)$ may be positive.*

Idea. The optimal expected cost $V(b)$ is concave in the belief, so splitting the belief by a probe outcome can only lower it on average; if the best action never changes, V is linear on the relevant beliefs and the split gains nothing (Fig. S1b,c).

Proof. Step 1. $V(b) = \min_a \mathbb{E}_b[J_\theta(a)]$ is a minimum of functions that are linear in b and is therefore concave. Since $\mathbb{E}_o[b_o] = b$, Jensen’s inequality gives $\mathbb{E}_o[V(b_o)] \leq V(b)$, so $\text{VOI}(p) \geq 0$.

Step 2. If $a^\dagger$ minimizes J_θ for b -almost every θ , every posterior b_o is supported on the same set, so $a^\dagger$ also minimizes $\mathbb{E}_{b_o}[J_\theta(\cdot)]$. Hence $\mathbb{E}_o[V(b_o)] = \mathbb{E}_o \mathbb{E}_{b_o}[J_\theta(a^\dagger)] = \mathbb{E}_b[J_\theta(a^\dagger)] = V(b)$.

Step 3. The mutual information $I(\theta; o \mid p)$ is positive whenever two values of θ with positive probability induce different outcome distributions, which is compatible with Step 2. $\square$

Consequently, information-maximizing probing probes whenever plausible contexts predict different outcomes, whereas value-of-information probing requires that they also prescribe different actions. When one memory dominates the uncertainty, we use a cheaper local criterion: for a single row with covariance Σ , gradient g of the cost difference between the two best actions, and probe features ϕ_p , a probe reduces the variance of that difference by $\Delta_p = (g^\top \Sigma \phi_p)^2 / (\phi_p^\top \Sigma \phi_p + \sigma_i^2(z_p, a_p))$. This linearized score is not the exact value of information, and we evaluate both.

B.7 Long-run latent safety (Proposition 4)

Proposition 4 (Long-run latent barrier violations). *Assume Eq. (7) is enforced exactly whenever it is feasible and that a fallback action otherwise satisfies the one-step barrier condition. Let scores lie in $[0, \bar{S}]$, margins be initialized in $[0, \bar{S}]$, $\gamma > 0$ and $0 < \delta_s < 1$, and let K_T count the memories created in the first T steps. Then for every sequence of contexts and cues,*

$$\sum_{t < T} \mathbf{1}\{h(z_{t+1}) < (1 - \alpha)h(z_t)\} \leq \delta_s T + K_T(\bar{S} + 2\gamma)/\gamma. \quad (\text{S1})$$

Idea. A violation can only happen on a step where the attributed memory’s margin was too small (a “miss”). Each margin follows the adaptive conformal update, which by a deterministic counting argument cannot miss much more than a fraction δ_s of its steps.

Proof. Step 1 (violations are misses). Suppose some plausible memory c covers step t , i.e. $s_{c,t} \leq b_{c,t}$, and Eq. (7) was enforced. By Lipschitz continuity of h ,

$$h(z_{t+1}) \geq h(\bar{z}'_{c,t}) - L_h s_{c,t} \geq h(\bar{z}'_{c,t}) - L_h b_{c,t} \geq (1 - \alpha)h(z_t).$$

If no action satisfied Eq. (7), the fallback gives the same conclusion. A violation can therefore only occur when no plausible memory covers the step, and by the attribution rule this step counts as a miss for its attributed memory.

Step 2 (margins stay bounded). Fix memory c . If $b_c > \bar{S}$, no miss is possible because scores are at most $\bar{S}$, so the update decreases b_c . Starting from $b_c \leq \bar{S}$, one update increases b_c by at most γ , hence $b_c \leq \bar{S} + \gamma$ at all times. Symmetrically, if $b_c < 0$ every step is a miss and b_c increases, so $b_c \geq -\gamma\delta_s$.

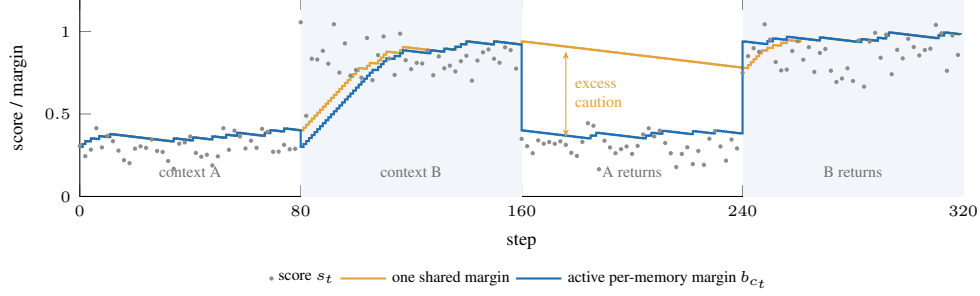


Figure S3: **Margins are remembered with memories** (synthetic simulation of the update in [Section 4](#), $\gamma = 0.02$, $\delta_s = 0.1$). Scores are drawn from a low-noise context A and a high-noise context B that alternate every 80 steps. A single shared margin grows during B and then decays slowly after A returns, so the filter stays over-cautious. With one margin per memory, the margin of each context is restored as soon as the context is recognized, and neither stream loses its calibration, as [Proposition 4](#) requires.

Step 3 (counting misses). Let memory c have T_c attributed steps and n_c misses. Summing the updates gives $b_{c,\text{end}} - b_{c,\text{start}} = \gamma(n_c - \delta_s T_c)$. By Step 2 the left side is at most $\bar{S} + \gamma + \gamma\delta_s$, so $n_c \leq \delta_s T_c + (\bar{S} + 2\gamma)/\gamma$.

Step 4. Summing over the K_T memories and using $\sum_c T_c = T$ together with Step 1 gives [Eq. \(S1\)](#). $\square$

The argument makes no assumption on the data, which is why it holds under misleading cues. A long-run violation rate of δ_s requires $K_T/T \rightarrow 0$, and resetting a margin costs as much as creating a new memory. [Figure S3](#) contrasts per-memory margins with a single shared margin on a synthetic stream. Physical safety additionally requires that h represents the physical constraint [\[3\]](#).

B.8 Recognition time

For a stored true context c and one alternative c' , let $\ell_0 = \log \frac{p(c|D_{<e})}{p(c'|D_{<e})} + \log \frac{p_\eta(y_e|c)}{p_\eta(y_e|c')}$ be the pre-contact log odds.

Proposition 5 (Two-context recognition time). *Assume fixed likelihoods and i.i.d. diagnostic observations under c whose log-likelihood-ratio increments X_s have mean $D > 0$. Let T be the first time the log odds $\ell_0 + \sum_{s \leq t} X_s$ reach $\ell_\star$, with $\mathbb{E}[T] < \infty$ and integrable overshoot $O_T \geq 0$. If $\ell_0 \geq \ell_\star$ then $T = 0$; otherwise $\mathbb{E}[T] = (\ell_\star - \ell_0 + \mathbb{E}[O_T])/D \geq (\ell_\star - \ell_0)/D$.*

Idea. The log odds form a random walk with drift D that starts at the cue-informed value ℓ_0 ; the expected time to climb the remaining distance $\ell_\star - \ell_0$ is that distance divided by the drift.

Proof. If $\ell_0 \geq \ell_\star$, the threshold is met before any observation and $T = 0$. Otherwise, by definition of T and the overshoot, $\sum_{s \leq T} X_s = \ell_\star - \ell_0 + O_T$. Since T is a stopping time with $\mathbb{E}[T] < \infty$ and the X_s are i.i.d. with mean D , Wald's identity gives $\mathbb{E}[\sum_{s \leq T} X_s] = D \mathbb{E}[T]$. Combining the two expressions and using $O_T \geq 0$ proves the claim. $\square$

A stronger cue raises ℓ_0 and shortens recognition, and better-separated memories raise D . Repeated exposure can improve both, but occurrence counts alone need not produce savings when competing contexts recur equally often. In a controlled two-context test with fixed memories, observed recognition times match [Proposition 5](#) within **XX%** ([Fig. S4](#)).

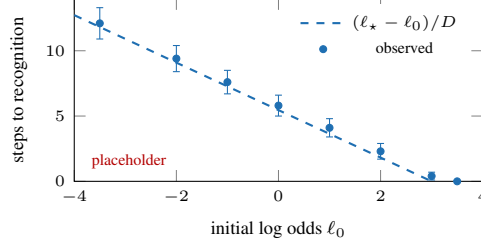


Figure S4: Recognition time against initial log odds in a controlled two-context test with fixed memories (placeholder data). The dashed line is the Wald prediction with the measured increment mean D and threshold $\ell_* = 3$; observed times include overshoot.

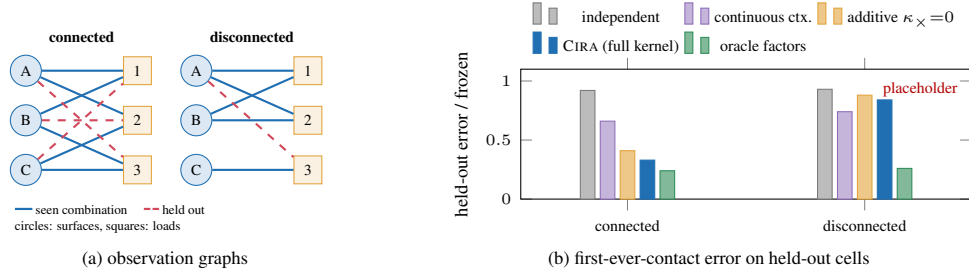


Figure S5: **Composition before the first-ever contact** (placeholder data in b). (a) Surfaces A–C and loads 1–3. In the connected design every held-out cell is linked to the seen cells by a path. In the disconnected control, the held-out cell A3 joins two components. (b) Prediction error at the first contact of each held-out combination. The factor prior transfers only in the connected design, as Proposition 2 predicts; the interaction term ($\kappa_{\times} > 0$) helps when surface and load interact.

C Additional Results

Figure S5 shows the factorial holdout design and the held-out error of all compositional baselines, including a continuous-context model with a state–action kernel multiplied by a context-descriptor kernel.

With predictive weights, the intervals of Theorem 1 retain their nominal coverage as contexts become harder to distinguish, whereas coverage drops to **XX%** with filtered weights and further with hard assignments (Fig. S6a), consistent with Remark 1. At a matched latent violation rate of **2%**, per-memory margins intervene **XX%** less often than a single adaptive margin and **XX%** less often than a filter that guards against all memories (Fig. S6b). Cumulative violations remained within Eq. (S1) in every run that satisfied its assumptions, including runs with misleading cues. Finally, value-of-information probing matches the success rate of information-maximizing probing with **XX%** fewer probes, because it abstains whenever all plausible contexts prescribe the same strike (Fig. S6c).

D Implementation Details

Component	Setting	Value
Correction subspace	principal directions of training residuals	$k = 8$
Features	$\phi(z, a) = [a, U^T z, 1]$	$p = 15$
Noise model	MLP on (z, a) , Gaussian NLL on training residuals	2×128
Context prior	sticky CRP: concentration, self-transition	$0.5, 0.95$
Factor kernel	$\kappa_0, \kappa_{\text{surf}}, \kappa_{\text{load}}, \kappa_{\times}$	$0.3, 1.0, 0.8, 0.4$
Memory creation	new-context hypothesis most probable for	5 steps
Cue model	Gaussian on PCA of region features	4 dims per factor
Planner	CEM: samples, elites, iterations, horizon	300, 30, 5, 10
Safety	$\alpha, \delta, \delta_s, \gamma$	$0.2, 0.05, 0.02, 0.01$

Table S3: Hyperparameters (placeholder values), fixed on development streams.

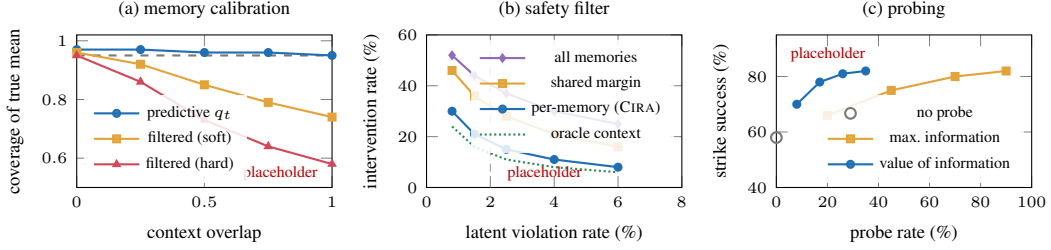


Figure S6: **Calibration, safety and probing** (placeholder data). (a) Coverage of the memory intervals of [Theorem 1](#) with oracle contamination terms; only predictive weights keep nominal coverage as contexts overlap ([Remark 1](#)). (b) Interventions against observed latent barrier violations over operating points; dashed line: the matched rate reported in the text. (c) Striking success against how often the robot probes first; value-of-information probing reaches the same success with fewer probes ([Proposition 3](#)).

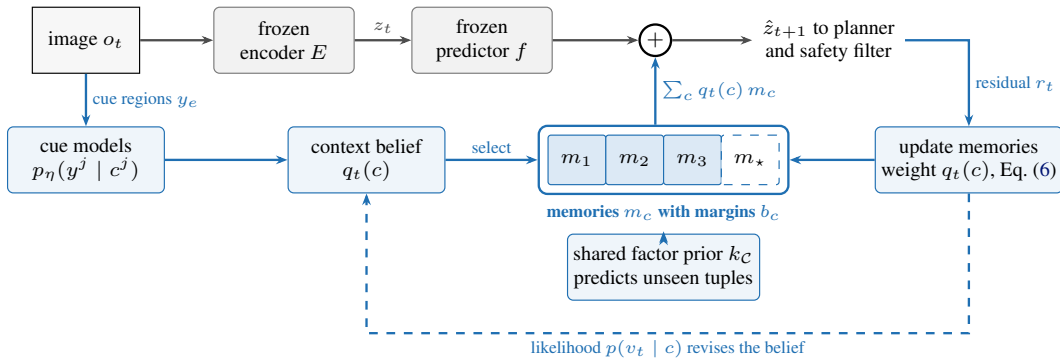


Figure S7: **CIRA overview.** A frozen world model predicts the next latent state, and a library of residual memories adds a correction weighted by the context belief. Before contact the belief comes from deployment history and learned cue models; after contact, dynamics likelihoods revise it. Each memory learns with the probability it had *before* the current error (q_t), stores its own safety margin, and shares a factor prior that predicts unseen combinations.

Streams. Development and evaluation streams use disjoint seeds and schedules. The cue channel, dwell-time distribution and held-out combinations are fixed before evaluation. Oracle methods receive true context labels only for selection, never extra transitions or perfect memory parameters.

Decomposition. Memory slots are aligned across snapshots by identity. A slot created after the pre-cue snapshot enters $\Theta^{(0)}$ with its data-independent creation prior and zero belief. Returns are analyzed only when the library membership is fixed, and creation windows are reported separately.

E Deployment Timeline

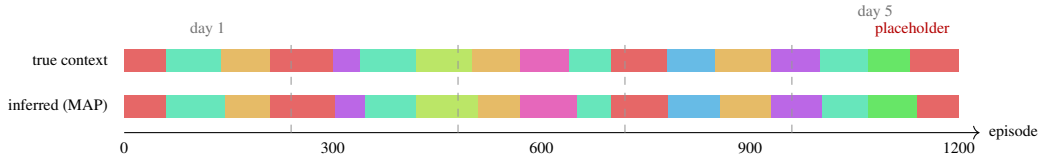


Figure S8: True and inferred contexts over the hardware deployment (placeholder schedule). Colours index the nine surface-load conditions; dashed lines mark day boundaries. Inferred switches lag true switches by the recognition time after contact, except where the cue alone identified the new condition.

References

- [1] Y. Abbasi-Yadkori, D. Pál, and C. Szepesvári. Improved algorithms for linear stochastic bandits. In *Advances in Neural Information Processing Systems*, volume 24, 2011. URL https://papers.nips.cc/paper_files/paper/2011/hash/e1d5be1c7f2f456670de3d53c7b54f4a-Abstract.html.
- [2] R. E. Krichevsky and V. K. Trofimov. The performance of universal encoding. *IEEE Transactions on Information Theory*, 27(2):199–207, 1981. doi:10.1109/TIT.1981.1056331.
- [3] P. Lutkus, K. Wang, L. Lindemann, and S. Tu. Latent representations for control design with provable stability and safety guarantees. In *IEEE Conference on Decision and Control*, 2025. URL <https://arxiv.org/abs/2505.23210>.